

Theory Week 10

n.ethz.ch/~bschen

Orga

Mechanics 3 CAB G59
WhatsApp group



10	18/11/24	Euler equations, examples	3.4.4
	20/11/24	spinning tops, gyroscopes	3.5
exercise 10: rotating-frame kinetics, Euler equations, gyroscopes			

AMB

$$\dot{H}_B^C = \dot{H}_B^M + \Omega^M \times H_B \quad \text{with} \quad H_B = I_B \omega + M(r_{CM} - r_B) \times v_B \quad (3.353)$$

If we choose point B such that $B = CM$ or $v_B = 0$, then

$$H_B = I_B \omega \quad \Rightarrow \quad \dot{H}_B^C = (I_B \omega)^{\circ M} + \Omega^M \times I_B \omega \quad (3.354)$$

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□ $\Rightarrow$ AMB in moving frame:

balance of angular momentum in a moving frame $\mathcal{M}$:

$$M_B = (I_B \omega)^{\circ M} + \Omega^M \times I_B \omega \quad \text{if} \quad \underline{B = CM} \quad \text{or} \quad \underline{v_B = 0}$$

□ If the $\mathcal{M}$ -frame rotates with the rigid body ($\Omega^M = \omega$)

$$\hookrightarrow M_B = (I_B \omega)^{\circ M} + \omega \times I_B \omega = I_B \dot{\omega}^M + \omega \times I_B \omega$$

since $\dot{I}_B^M = 0$

$$\Omega^M = \omega$$

□ if $\hat{M}$ aligns with principle axes of B $\Rightarrow$ Euler equations

body frame

Euler's equations in a rotating (principal) body frame $\hat{\mathcal{M}}$ ($\Omega^{\hat{\mathcal{M}}} = \omega$):

$$\left. \begin{aligned} \hat{I}_1 \dot{\omega}_1 + (\hat{I}_3 - \hat{I}_2) \omega_3 \omega_2 &= [M_{B,1}]_{\hat{\mathcal{M}}} \\ \hat{I}_2 \dot{\omega}_2 + (\hat{I}_1 - \hat{I}_3) \omega_1 \omega_3 &= [M_{B,2}]_{\hat{\mathcal{M}}} \\ \hat{I}_3 \dot{\omega}_3 + (\hat{I}_2 - \hat{I}_1) \omega_2 \omega_1 &= [M_{B,3}]_{\hat{\mathcal{M}}} \end{aligned} \right\} \quad \text{if} \quad \Omega^M = \omega \quad \text{and} \quad \begin{aligned} \omega_i &= [\omega_i]_{\hat{\mathcal{M}}} \\ \dot{\omega}_i &= [\dot{\omega}_i]_{\hat{\mathcal{M}}} = \frac{d}{dt} [\omega_i]_{\hat{\mathcal{M}}} \end{aligned}$$

$\neq 0$ unless $\omega = \text{const}$

here I_B is diagonal. By then just applying this property & $\Omega^M = \omega$, we derive the Euler eq.

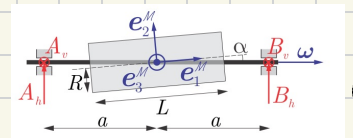
□ if Body = axisymmetric (moment of inertia does not change when rotating about axis of rot. sym.)

$\hookrightarrow$ introduce moving frame $\mathcal{B}$ aligned with princ. axes but $\Omega^M \neq \omega$

angular momentum balance in a rotating (principal) frame $\mathcal{B}$ if $\dot{I} = 0$ ($\Omega^{\mathcal{B}} \neq \omega$):

$$\left. \begin{aligned} \hat{I}_1 \dot{\omega}_1 + \hat{I}_3 \Omega_2^{\mathcal{B}} \omega_3 - \hat{I}_2 \Omega_3^{\mathcal{B}} \omega_2 &= [M_{B,1}]_{\mathcal{B}} \\ \hat{I}_2 \dot{\omega}_2 + \hat{I}_1 \Omega_3^{\mathcal{B}} \omega_1 - \hat{I}_3 \Omega_1^{\mathcal{B}} \omega_3 &= [M_{B,2}]_{\mathcal{B}} \\ \hat{I}_3 \dot{\omega}_3 + \hat{I}_2 \Omega_1^{\mathcal{B}} \omega_2 - \hat{I}_1 \Omega_2^{\mathcal{B}} \omega_1 &= [M_{B,3}]_{\mathcal{B}} \end{aligned} \right\} \quad \text{where} \quad \begin{aligned} \omega_i &= [\omega_i]_{\mathcal{B}} \\ \dot{\omega}_i &= [\dot{\omega}_i]_{\mathcal{B}} = \frac{d}{dt} [\omega_i]_{\mathcal{B}} \end{aligned}$$

e.g. poorly aligned rotating shaft



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Use Euler whenever you can !!!

Let us point out a common misconception. We introduced the following notation to compute time derivatives with respect to moving and non-moving frames, which also applies to ω :

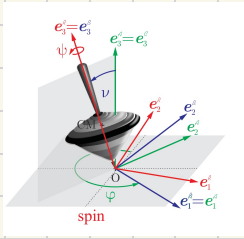
$$\omega(t) = \sum_{i=1}^3 \omega_i(t) e_i^M(t) \quad \Rightarrow \quad \dot{\omega} = \sum_{i=1}^3 (\dot{\omega}_i e_i^M + \omega_i \dot{e}_i^M), \quad \dot{\omega}^M = \sum_{i=1}^3 \dot{\omega}_i e_i^M. \quad (3.359)$$

While $\dot{\omega}$ represents the **absolute angular acceleration** (with respect to an inertial frame), $\dot{\omega}^M$ is the **angular acceleration measured with respect to the moving frame**. The latter is the one which enters the Euler equations (3.358), and its components are $[\dot{\omega}_i]_{\mathcal{M}} = \dot{\omega}_i$. It is important to realize that, even though $\Omega^M = \omega$ (so the $\mathcal{M}$ -frame is rotating with the body), the time derivative $\dot{\omega}$ is **not necessarily zero**. When we say "measured in the moving frame" or "measured with respect to the moving frame", we mean a quantity measured with respect to the moving coordinate basis. For example, an object fixed on a rotating disk has a non-zero velocity $v = \dot{x}$ with respect to an inertial frame, but it has a vanishing velocity $\dot{x}$ with respect to the corotating coordinate system $\mathcal{M}$ (an observer sitting on the disk will not see any (relative) motion of the other points on the disk). However, even though the observer is rotating with $\mathcal{M}$, they will still feel a change in the angular velocity. Therefore, even with respect to the corotating observer, the angular velocity of the disk is changing and $\dot{\omega} \neq 0$ unless $\omega = \text{const}$.

3.5 Spinning tops & gyroscopes

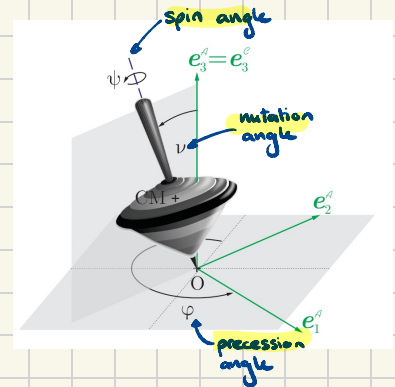
Spinning tops: $\text{DOF}_S = 6 - 3 = 3$

↑ no translation



$$\underline{\omega}_{\text{tot}} = \omega_{\varphi} + \omega_{\nu} + \omega_{\psi}$$

$$= \dot{\varphi} \underline{e}_3^a + \dot{\nu} \underline{e}_1^s + \dot{\psi} \underline{e}_3^s$$



↳ using R-frame we can use Euler eq.
! make sure to use $\hat{I}_{\text{spin}}$

| since aligned with princ. axes
& $\underline{v}_O = 0$

Def: steady precession : $\nu = \text{const}$, $\dot{\psi} = \text{const}$, $\dot{\varphi} = \text{const}$

→ usually found in fast spinning tops with M_{ext} only by gravity

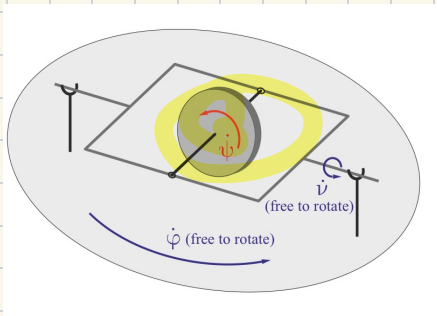
↳ if $|\dot{\psi}| \gg |\dot{\varphi}|$ (gyroscopic limit)

⇒ TSP-rule :

$$\dot{\varphi}_{\text{precession}} \times \dot{\psi}_{\text{spin}} \approx \frac{M_O}{\hat{I}_{\text{spin}}}$$

base right hand!

Gyroscopes:



A spinning wheel/disc mounted on a suspension, s.t. its spin axis can freely assume any orientation

↳ to reorient freely there must be no torques on the spinning body
⇒ no matter how you tilt/mount it, it will continue to spin around same axis

→ used in navigation